Project Brief: IMDB Movies Analysis using SQL

Objective

This project uses SQL to explore and analyse an IMDB movie database. It answers business-style questions about movies and directors, such as which movies are the most popular or profitable, and which directors are the most productive or bankable.

Database

The data sits in a MySQL database called project_movie_database and has two tables:

- **Directors:** director name, unique ID, gender (0/2 = male, 1 = female) and department.
- **Movies:** movie ID, titles, budget, revenue, popularity, release date, average rating, vote count, overview, tagline, UID and director ID.

The two tables are linked by joining directors.id with movies.director_id, so each movie can be matched to its director.

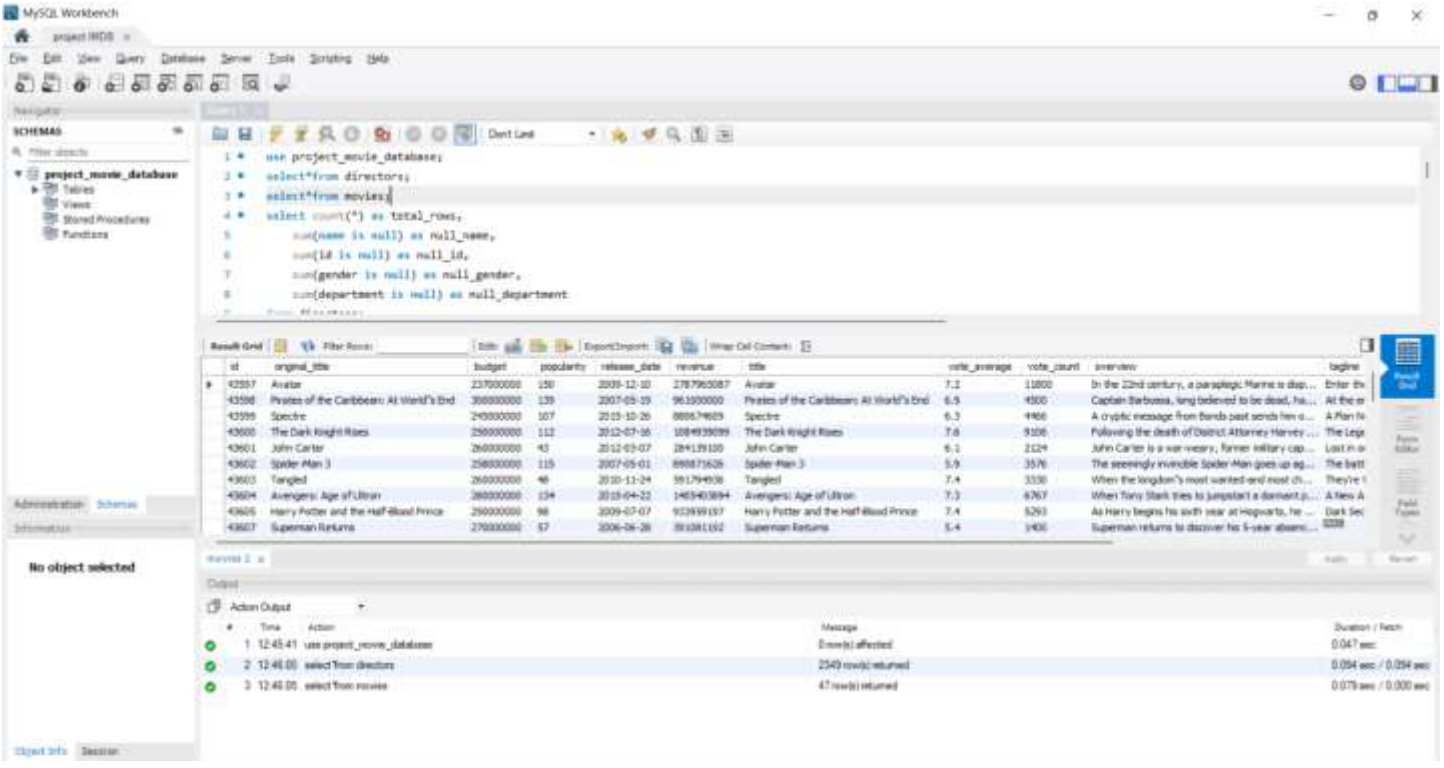
Approach

1. Explored both tables to understand their structure and size.
2. Checked data quality: null values, zero budget or revenue entries, blank text fields, and movies with no matching director.
3. Wrote queries to answer the problem statements, using these SQL concepts:
    - Filtering with WHERE, IN and LIKE
    - Aggregation with COUNT and SUM
    - Grouping with GROUP BY
    - Sorting and limiting with ORDER BY and LIMIT
    - Joining the two tables with JOIN
    - Calculated columns, such as profit = revenue – budget

Project Questions answered are as follows:-

a) Can you get all data about movies?

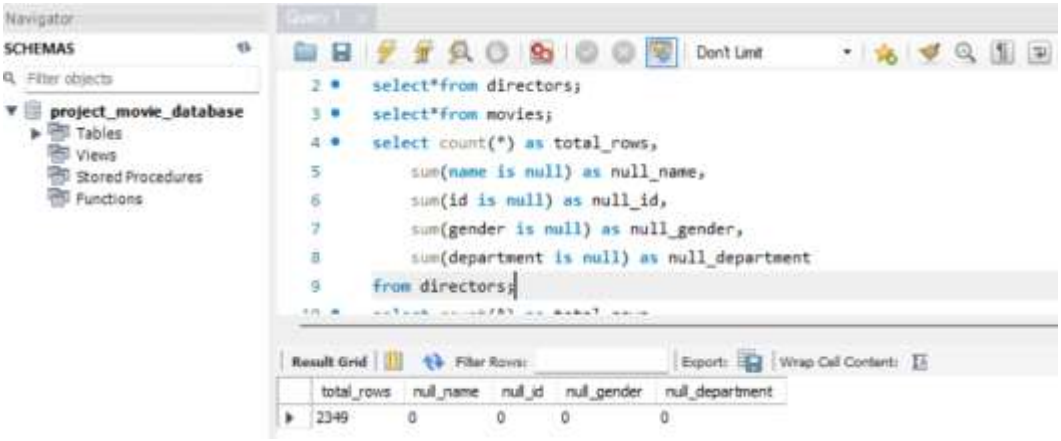
Ans:- use project_movie_database;
select*from directors;
select*from movies;

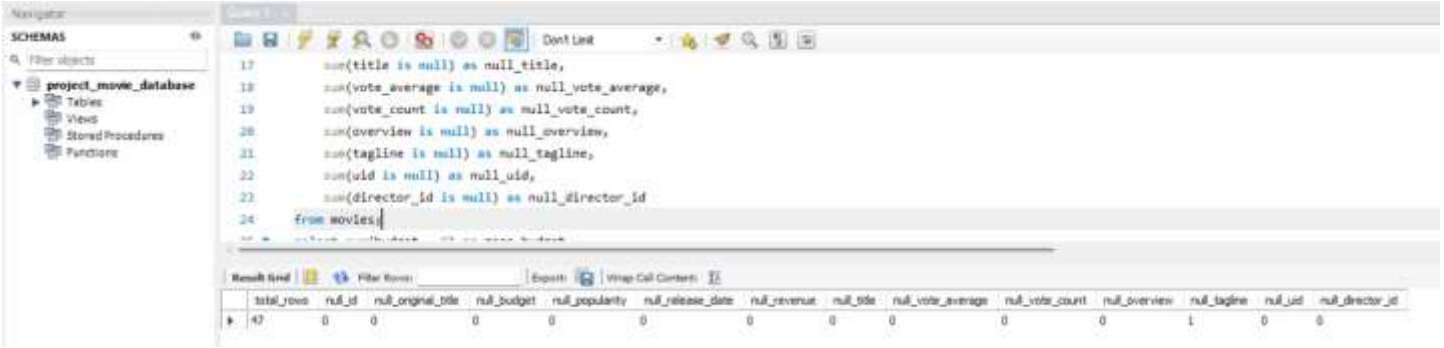


:- This Query Gives the Dataset for further Analysis

After Getting the data Firstly Check for Null Values in Both the Tables:

```
select count(*) as total_rows,
   sum(name is null) as null_name,
   sum(id is null) as null_id,
   sum(gender is null) as null_gender,
   sum(department is null) as null_department
from directors;
select count(*) as total_rows,
   sum(id is null) as null_id,
   sum(original_title is null) as null_original_title,
   sum(budget is null) as null_budget,
   sum(popularity is null) as null_popularity,
   sum(release_date is null) as null_release_date,
   sum(revenue is null) as null_revenue,
   sum(title is null) as null_title,
   sum(vote_average is null) as null_vote_average,
   sum(vote_count is null) as null_vote_count,
   sum(overview is null) as null_overview,
   sum(tagline is null) as null_tagline,
   sum(uid is null) as null_uid,
   sum(director_id is null) as null_director_id
from movies;
```

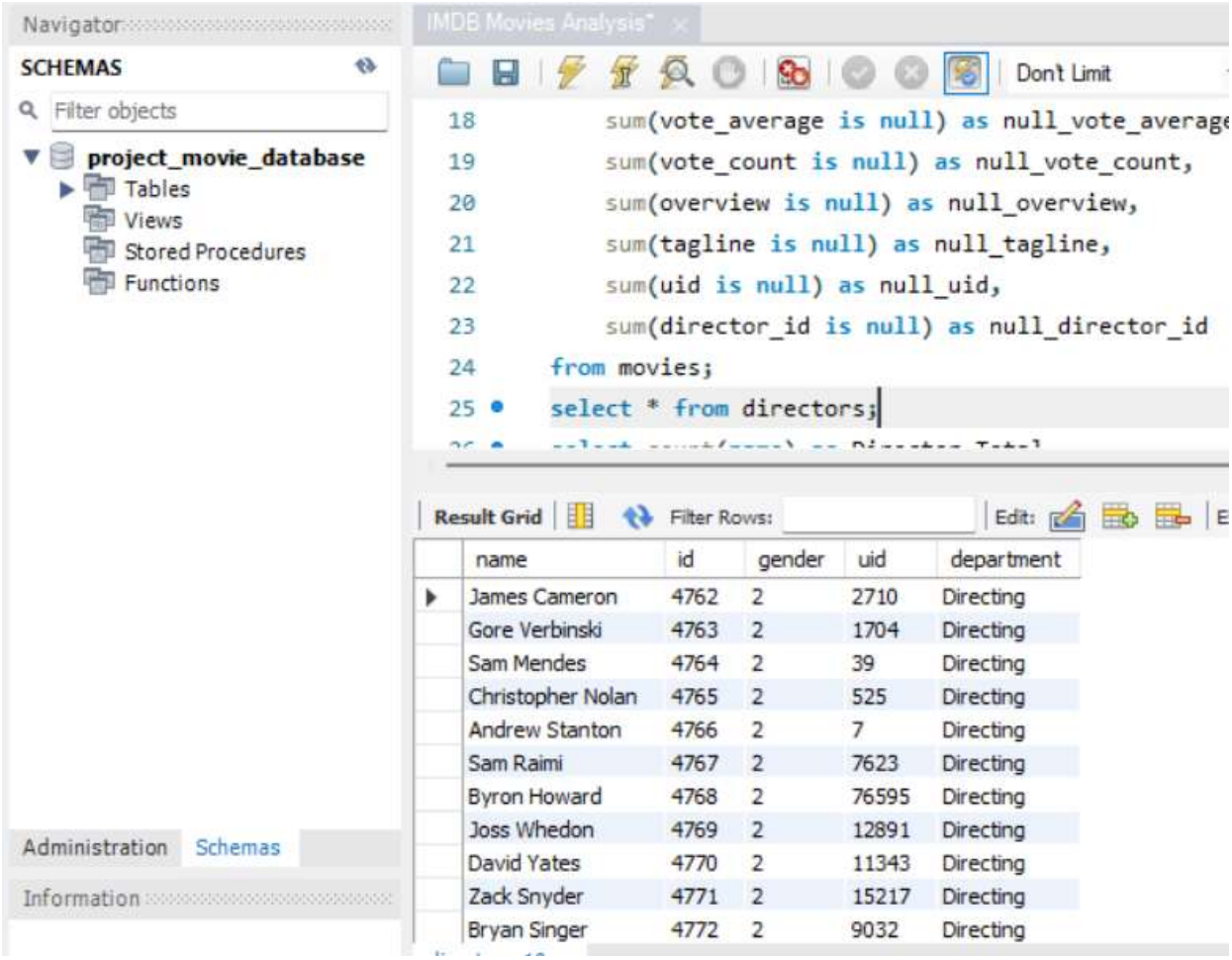




:- Null Values Checked and ignored not Important Null Values from both Tables.

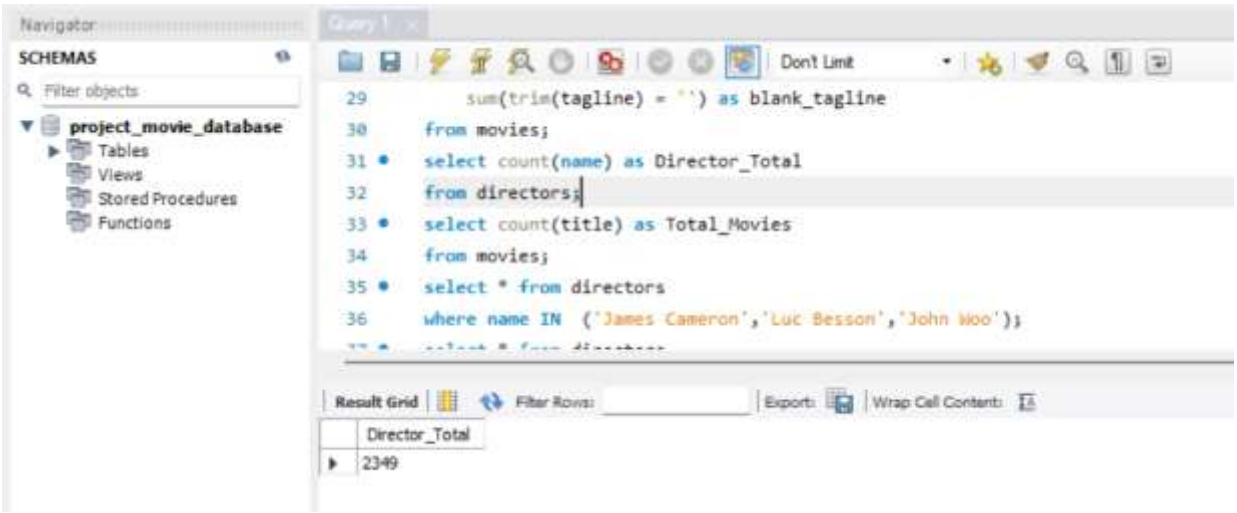
b) How do you get all data about directors?

Ans:- select * from directors;



:-This Query gives all data About Directors.

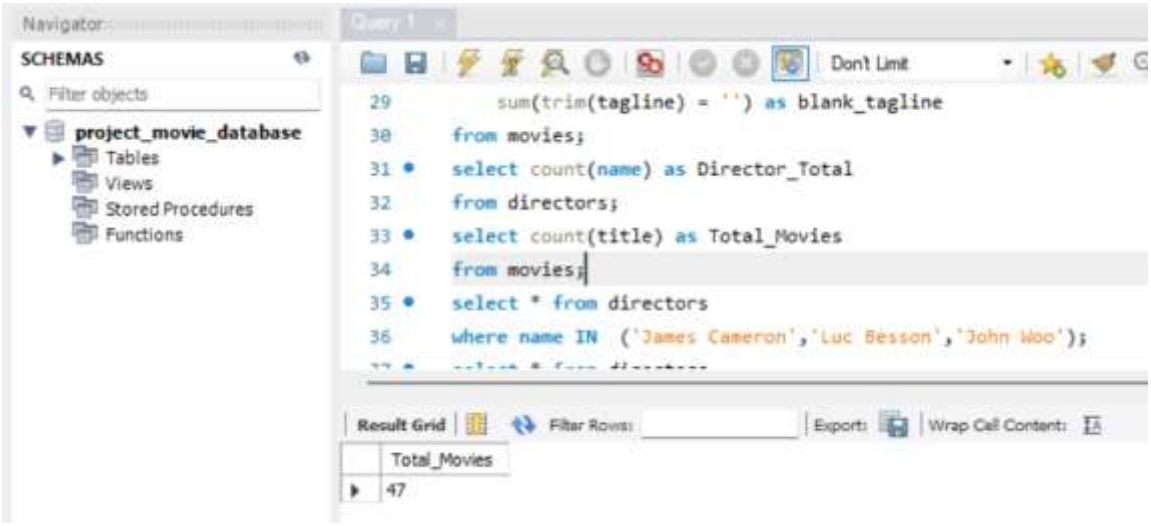
select count(name) as Director_Total
from directors;



:- This Query Gives Total count of Directors.

c) Check how many movies are present in IMDB.

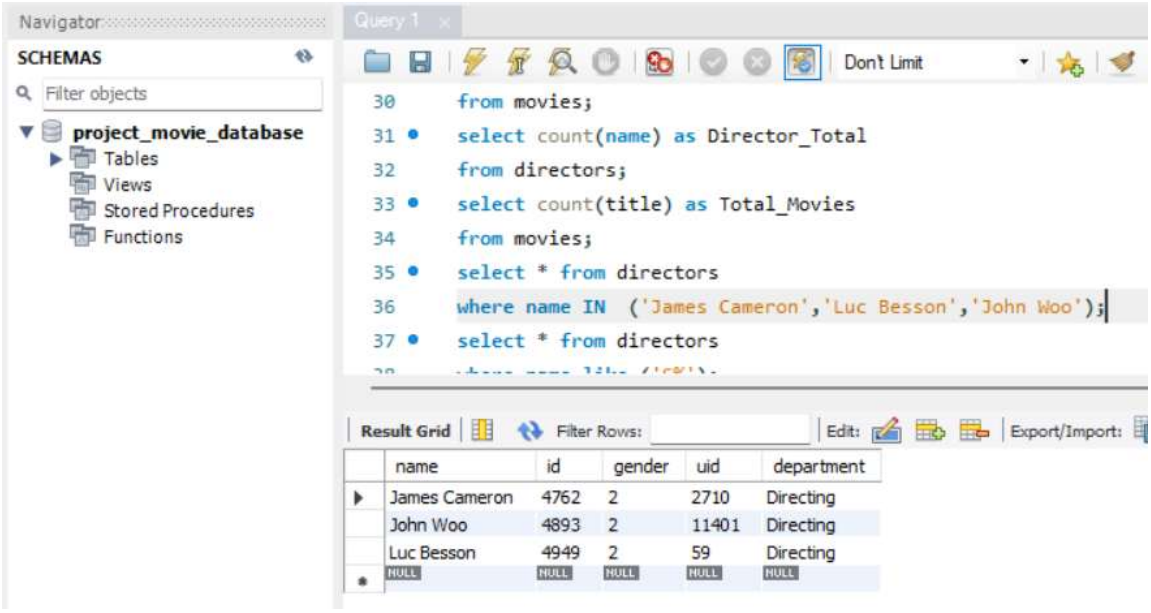
Ans:- select count(title) as Total_Movies
from movies;



:- This Query Gives the Total Count of Movies Present in the Movies Table.

d) Find these 3 directors: James Cameron ; Luc Besson ; John Woo

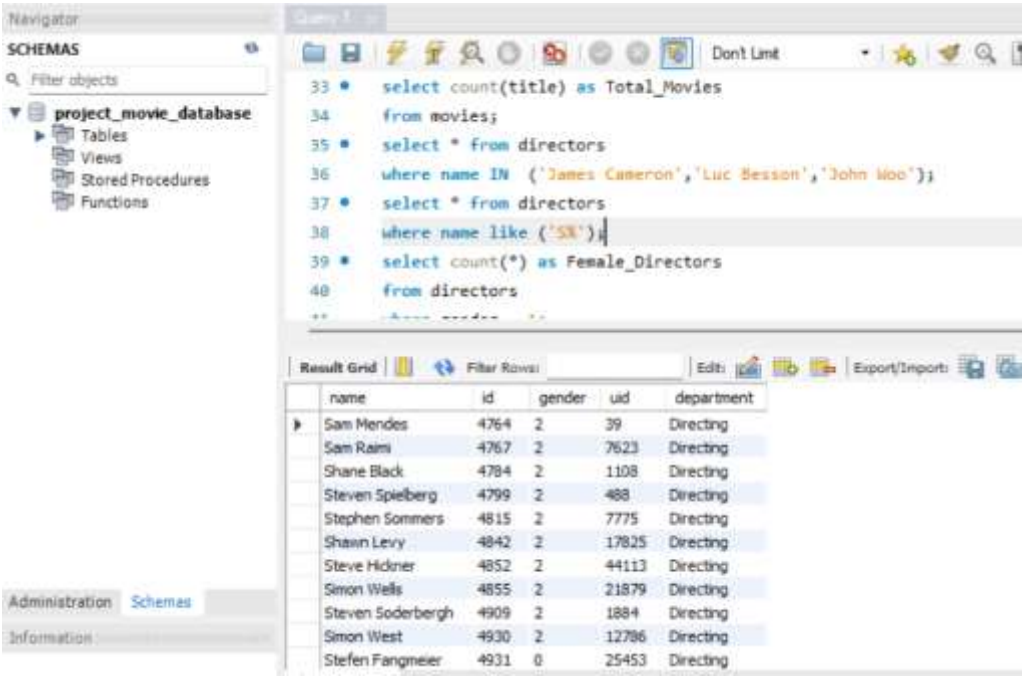
Ans:- select * from directors
where name IN ('James Cameron','Luc Besson','John Woo');



:-This Query Finds the Directors along with there details.

e) Find all directors with name starting with S.

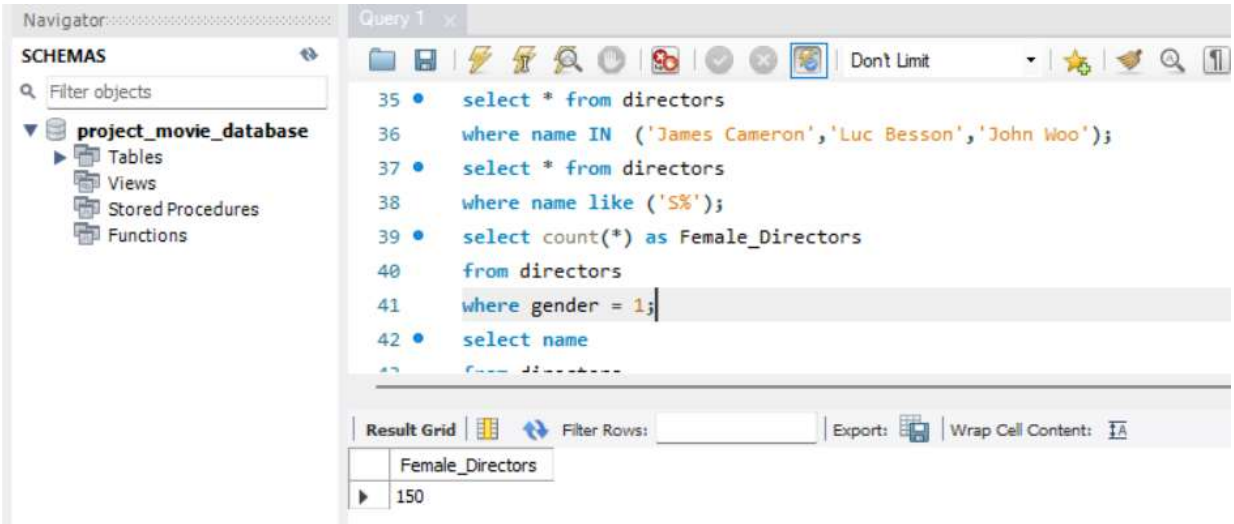
Ans:- select * from directors
where name like ('S%');



:-This Query Gives the Names of Directors starting from “s”.

f) Count female directors.

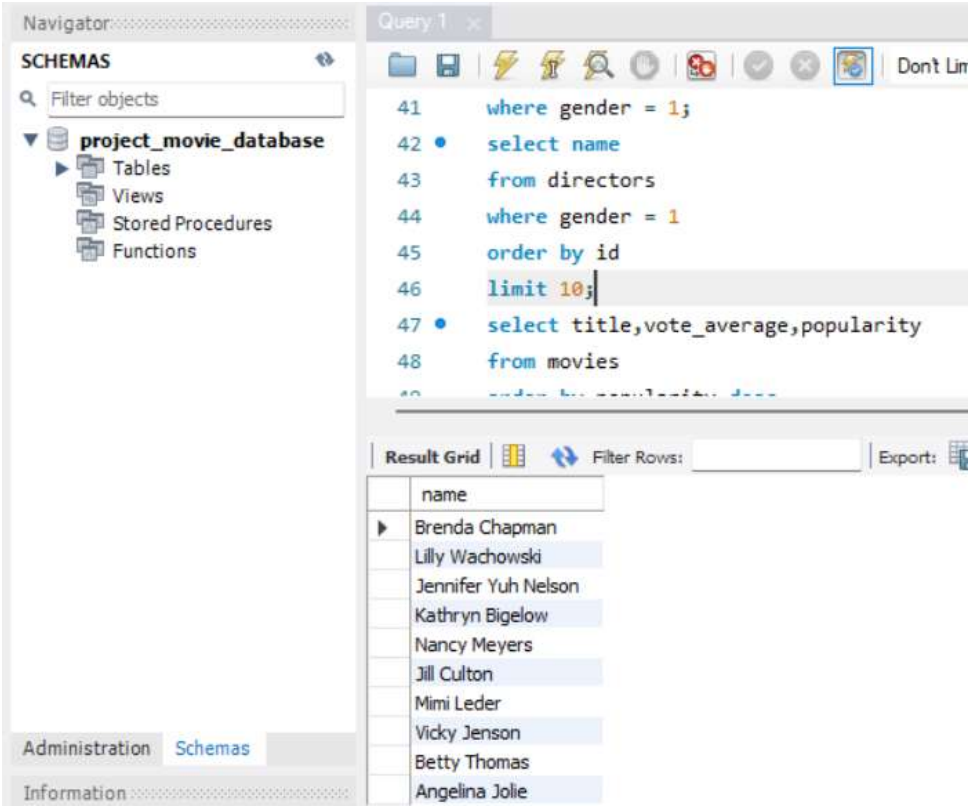
Ans:- select count(*) as Female_Directors
from directors
where gender = 1;



:-This Query Counts the Total number of Female Directors.

g) Find the name of the 10th first women directors?

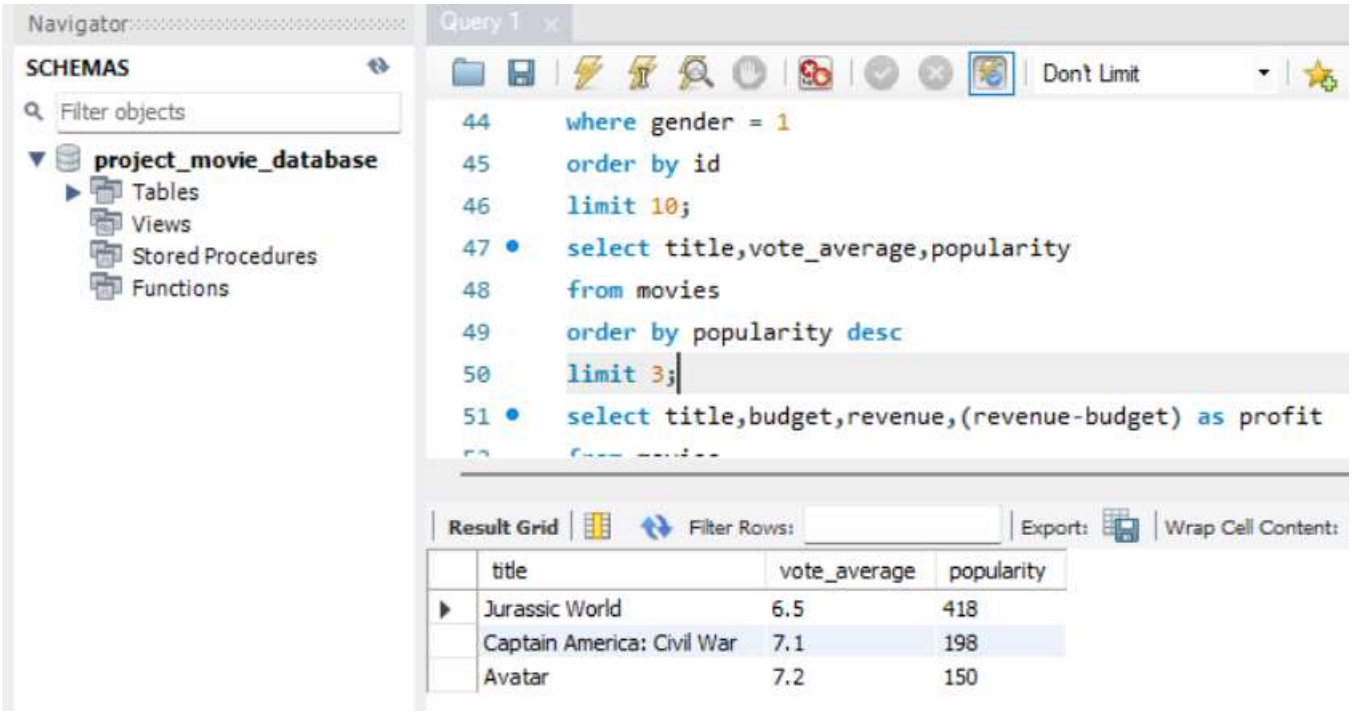
Ans:- select name
from directors
where gender = 1
order by id
limit 10;



:-This Query Gives the name of first 10 Female Directors from table directors by their ID.

h) What are the 3 most popular movies?

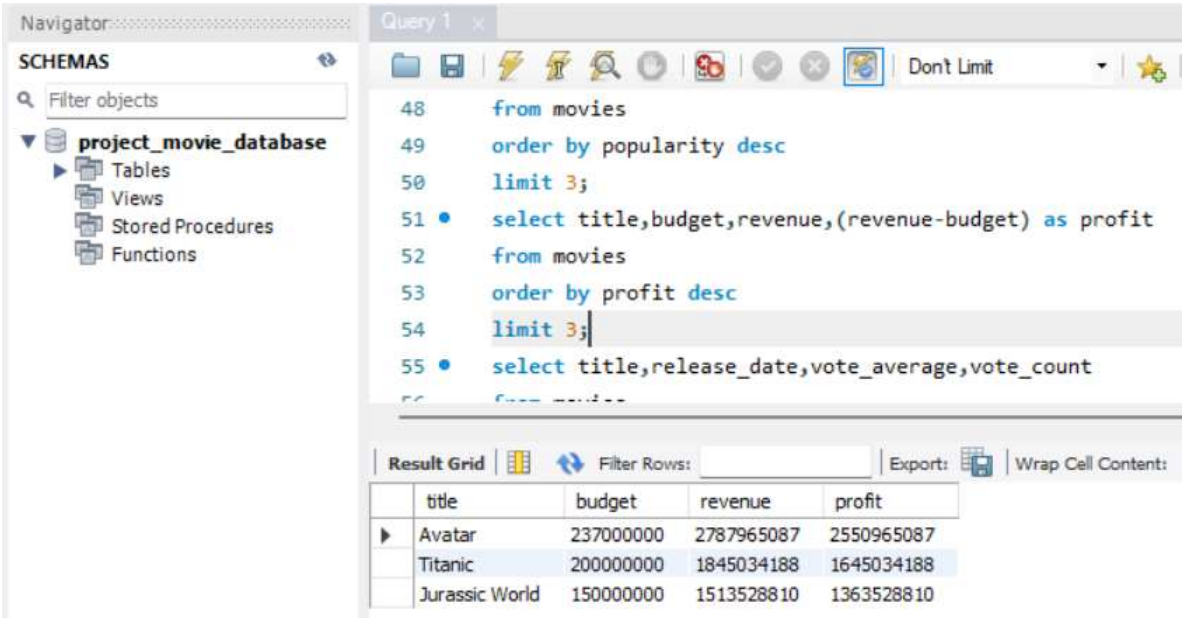
Ans:- select title,vote_average,popularity
from movies
order by popularity desc
limit 3;



:-This Query Gives the Three Most Popular Movies.

i) What are the 3 most bankable movies?

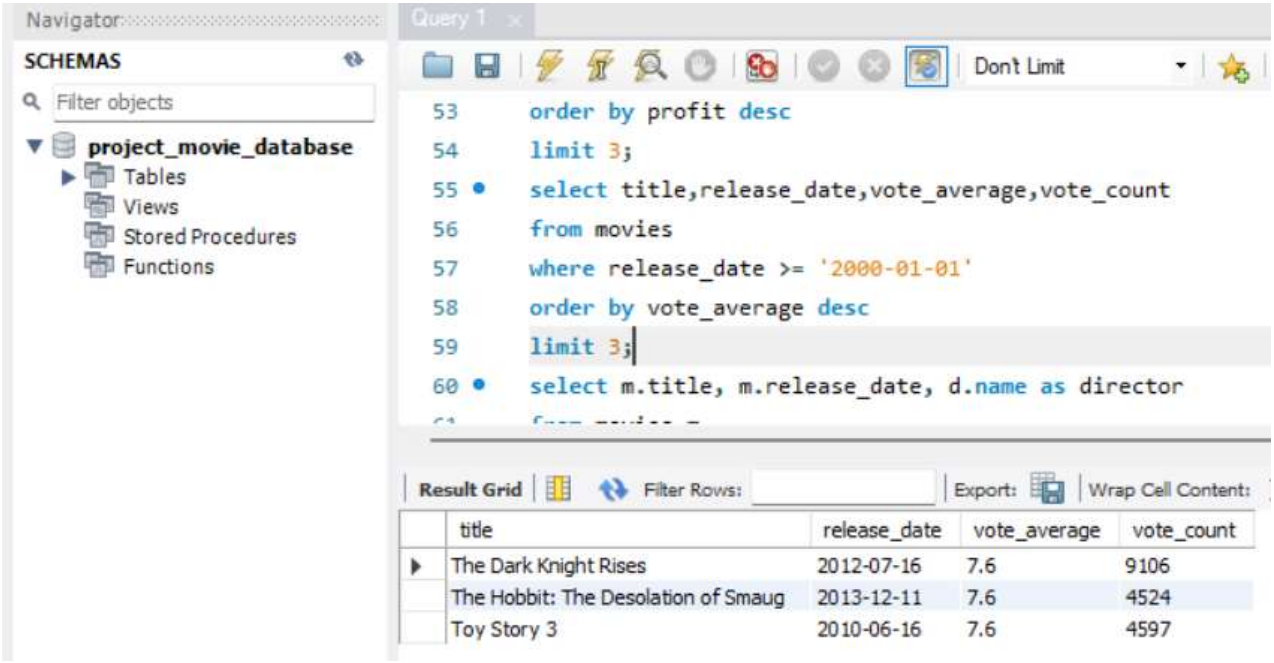
Ans:- select title,budget,revenue,(revenue-budget) as profit
from movies
order by profit desc
limit 3;



:-This Query Gives the Three most Profitable movies in the Movies Table.

j) What is the most awarded average vote since the January 1st, 2000?

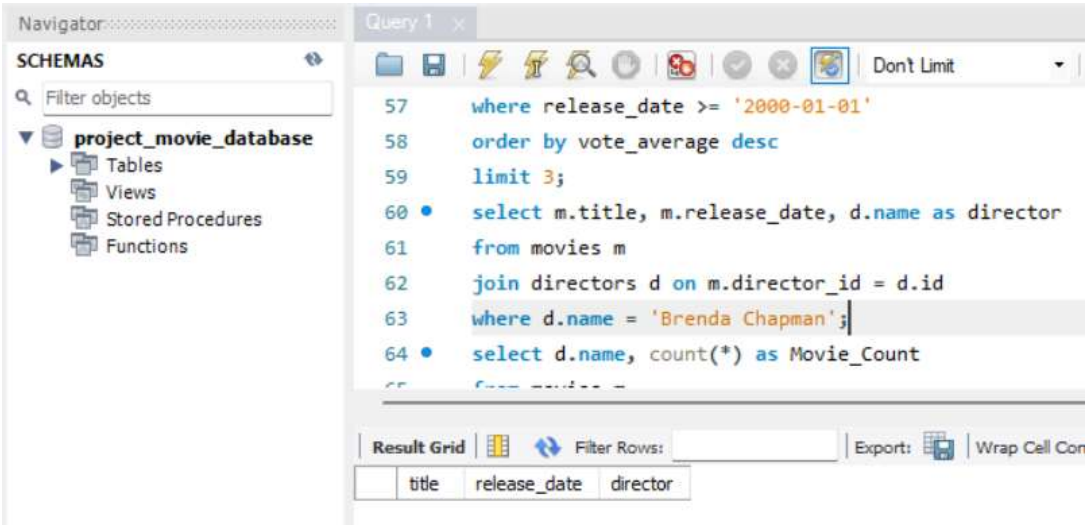
Ans:- select title,release_date,vote_average,vote_count
from movies
where release_date >= '2000-01-01'
order by vote_average desc
limit 3;



:-This Query sorts and gives the three most Voted Movies in the Movies table.

k) Which movie(s) were directed by Brenda Chapman?

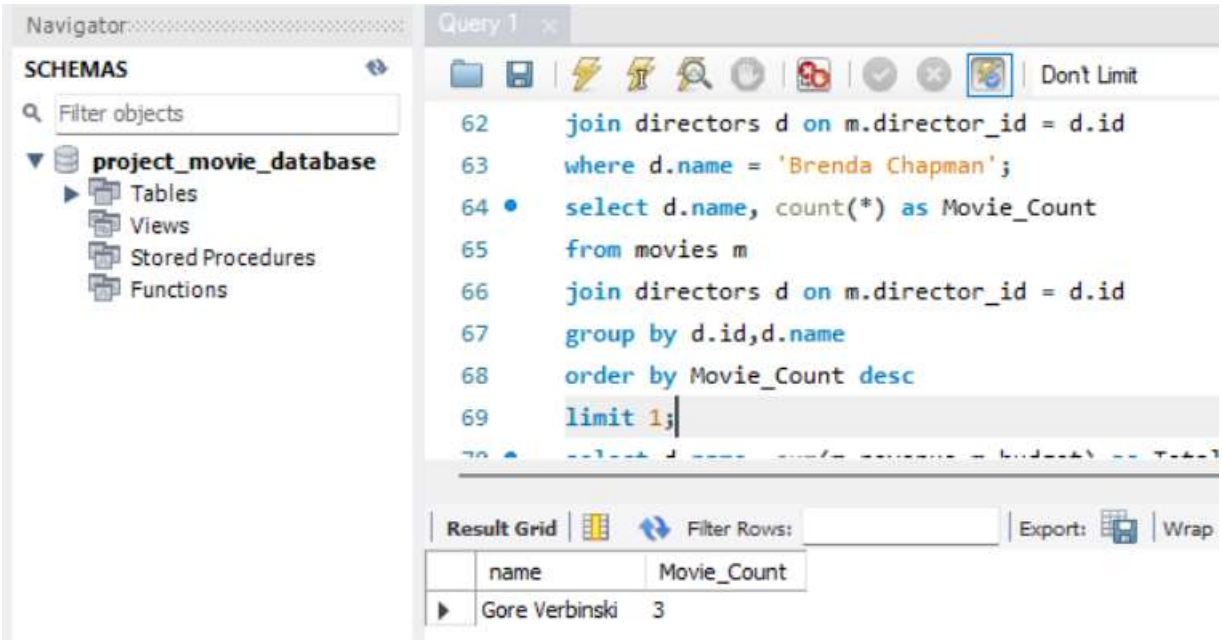
Ans:- select m.title, m.release_date, d.name as director
from movies m
join directors d on m.director_id = d.id
where d.name = 'Brenda Chapman';



:-This Query Gives the movies Directed by the Mentioned Director, as it has no Values means she has’nt Directed any movies in the data present in the table.

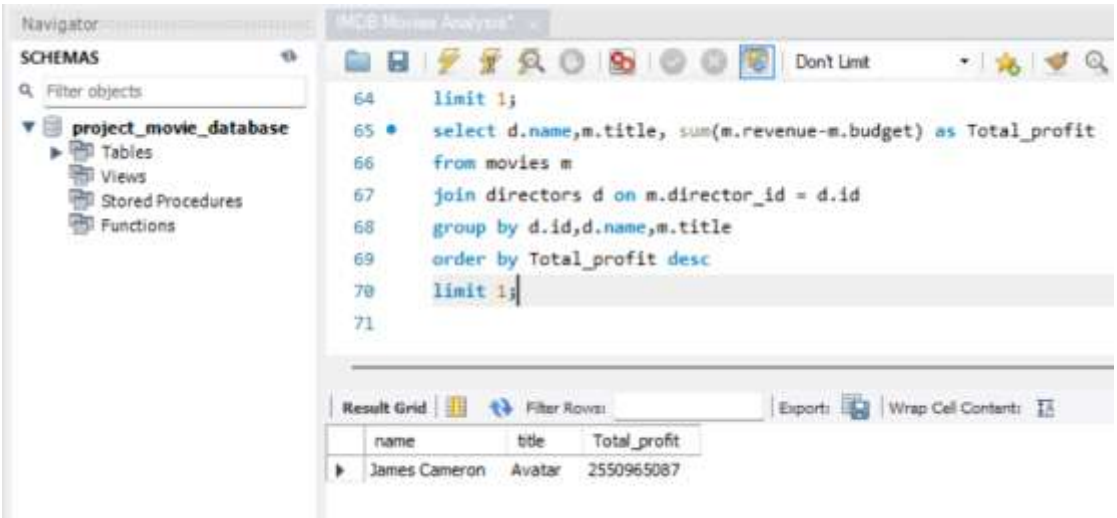
l) Which director made the most movies?

Ans:- select d.name, count(*) as Movie_Count
from movies m
join directors d on m.director_id = d.id
group by d.id,d.name
order by Movie_Count desc
limit 1;



:-This Query Gives the Name of the Director who has made the most movies in the Data present in the Table.

m) Which director is the most bankable?
n)
Ans:- select d.name, m.title, sum(m.revenue-m.budget) as Total_profit
from movies m
join directors d on m.director_id = d.id
group by d.id,d.name,m.title
order by Total_profit desc
limit 1;



:-This Query Gives the Name of the Director whose movie was most Profitable along with the movie name.

Key Questions Answered

- Total number of movies and details of specific directors (James Cameron, Luc Besson, John Woo)
- Directors whose names start with “S”, and the count of female directors
- The most popular movies and the most bankable (highest profit) movies
- The highest-rated movie released since 1 January 2000
- Movies directed by Brenda Chapman
- The director with the most movies and the most bankable director

Key Takeaway

The project shows how SQL can turn raw movie data into useful insights, from basic retrieval to joins and aggregations. It also stresses data cleaning, since missing or zero values in fields like budget and revenue can distort profitability results.

If you’d like this as a Word or PDF file, I can generate it. I can also add a short “Observations” section once you have your actual query results.

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